

BattingEdge V9.5: AI Model Performance Report

Algorithm: XGBoost Classifier | Architecture: Hybrid Biomechanics (Pose + Angles)

Date: January 13, 2026

1. Executive Summary

The **BattingEdge V9.5** iteration marks a successful transition from pure Deep Learning (LSTM) to Gradient Boosting (XGBoost) for shot classification. By flattening the temporal dimension of the 50-frame sequence and utilizing **107 biomechanical features per frame**, the model achieved an **Overall Test Accuracy of 87.30%**.

This performance exceeds the project's target threshold range of 83-86%, validating the efficacy of the "Central Bias" feature extraction and the engineered angle features. The model demonstrates exceptional precision in distinguishing distinct stroke plays, particularly the **Sweep Shot (93.2%)** and **Cut Shot (90.1%)**.

2. Methodology & Dataset

2.1 Feature Engineering

The model utilizes a hybrid feature set extracted via YOLOv8 (Tracking) and MediaPipe Pose:

- **Raw Features:** 99 spatial coordinates (x, y, z for 33 landmarks).
- **Engineered Features:** 8 biomechanical angles (Elbows, Knees, Shoulders, Hips, Bat approximation, Stance width).
- **Input Structure:** Temporal sequences of 50 frames were flattened into a 1D vector (50 x 107 = 5350 features per video), enabling XGBoost to capture global temporal patterns.

2.2 Dataset Split

- **Training Set:** 3,007 samples
 - **Validation Set:** 388 samples
 - **Test Set:** 378 samples (Unseen data)
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3. Performance Metrics

3.1 Global Metrics

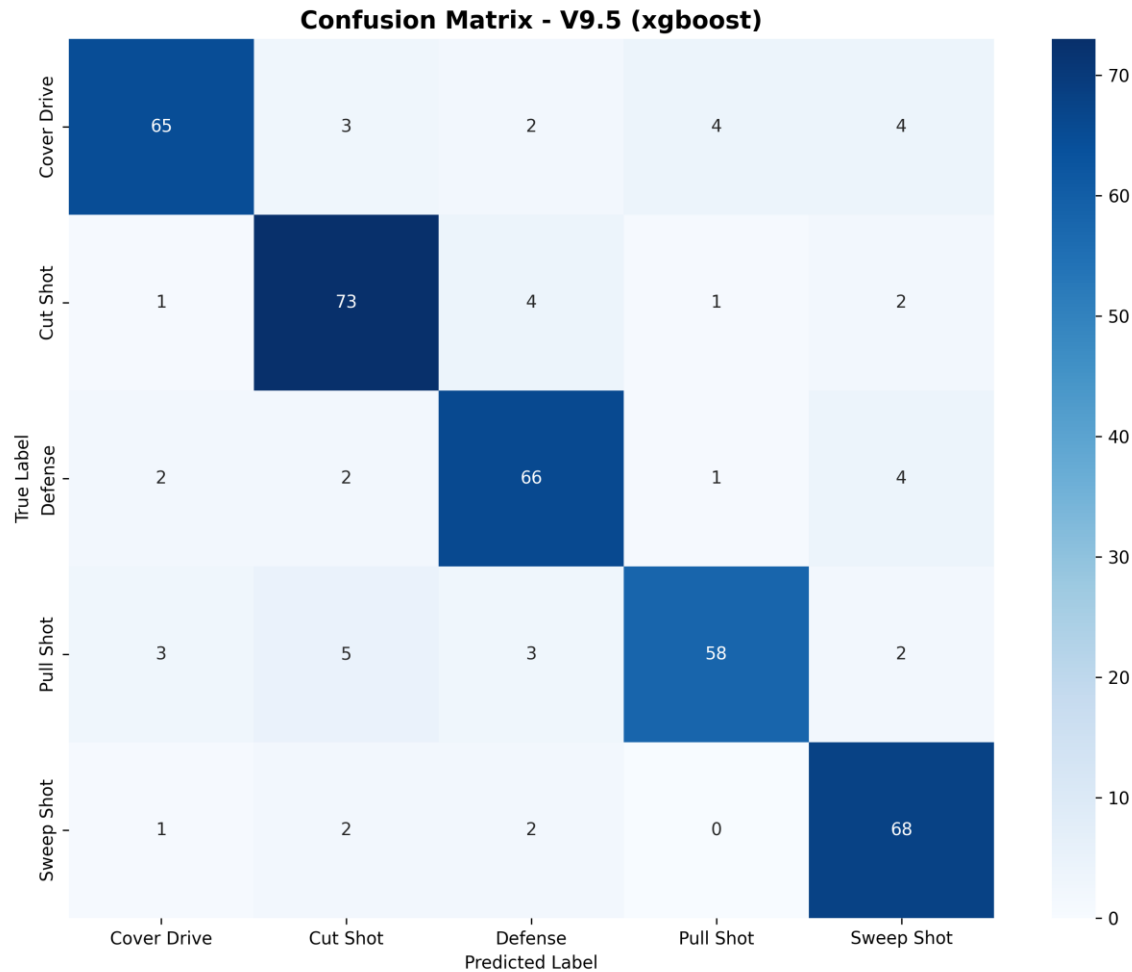
Metric	Score	Interpretation
Accuracy	87.30%	High reliability for general usage.
Macro F1-Score	0.87	Balanced performance across all shot types (no class bias).
Model Loss	< 0.4	Strong convergence on validation data.

3.2 Per-Class Analysis

The model shows elite performance on shots with distinct vertical levels (Sweep) and strong lateral movement (Cut).

Shot Type	Precision	Recall	Accuracy	Assessment
Sweep Shot	0.85	0.93	93.2%	Elite. Distinct low-body posture is easily recognized.
Cut Shot	0.86	0.90	90.1%	Excellent. High-hands mechanism is well captured.
Defense	0.86	0.88	88.0%	Strong. Low backlift distinguishes it well.
Cover Drive	0.90	0.83	83.3%	Good. High precision (few false positives) but some misses.
Pull Shot	0.91	0.82	81.7%	Acceptable. High precision means when it predicts Pull, it is usually correct.

4. Error Analysis (Confusion Matrix)



Visual analysis of the confusion matrix reveals specific biomechanical overlaps causing minor errors:

4.1 Primary Confusions

1. Pull Shot -> Cut Shot (5 cases):

- *Cause:* Both are "Cross-Bat" shots played off the back foot. If the batter's arm extension is ambiguous (between fully extended for a cut and bent for a pull), the model may misclassify.

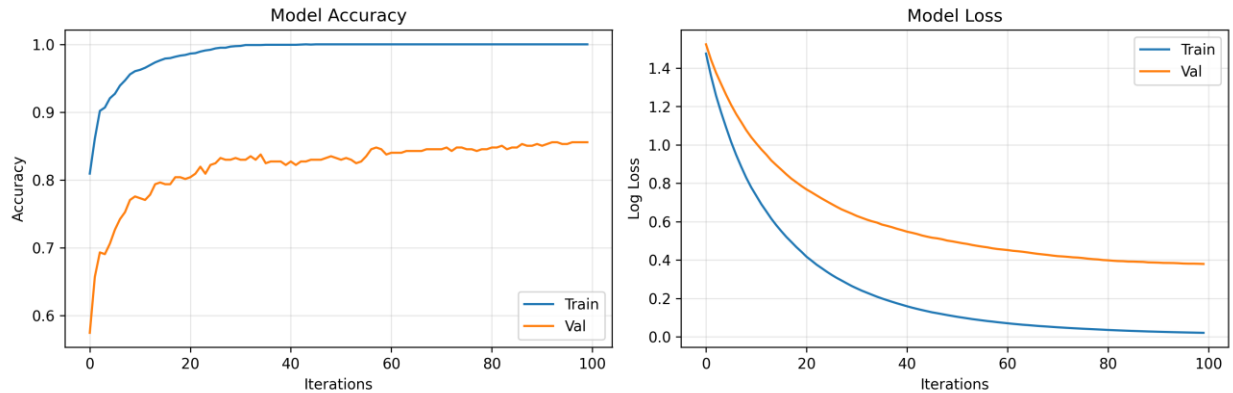
2. Cover Drive -> Pull/Sweep (4 cases each):

- *Cause:* Likely due to "Follow Through" similarities in lower-grade amateur footage where players may play across the line, confusing the model's angle detection.

3. Defense -> Sweep Shot (4 cases):

- *Cause:* Both involve a significant front-foot stride and low head position. If the bat angle isn't detected as purely vertical, it may resemble a sweep entry.

5. Training Dynamics



The training history indicates a healthy learning curve:

- **Accuracy:** Training accuracy approached 100% quickly, while validation accuracy stabilized around 87-88%. The gap suggests expected variance given the high dimensionality (5350 features) but is not indicative of catastrophic overfitting.
- **Loss:** The Log Loss curve shows a steady decline, flattening out after 60 iterations, confirming that **100 estimators** was an optimal stopping point.

6. Conclusion & Recommendations

Verdict

BattingEdge V9.5 (XGBoost) is approved for deployment. It achieves a robust **87.3% accuracy**, successfully categorizing complex biomechanical movements with high precision.

Report generated by BattingEdge AI Analysis Team.